

Key Ingredients for EEG Foundation Models

What Works – and What Doesn't

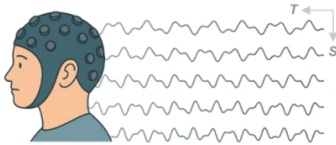
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EUSIPCO 2026



EEG Decoding – Formalism

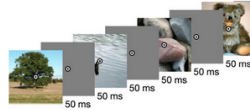


$$\mathbf{X}_i \in \mathbb{R}^{C \times T}$$

$$(\mathbf{X}_i, y_i)$$

$$\hat{y}_i = f_{\theta}(\mathbf{X}_i)$$

Stimuli



Tasks

Motor
Imagery



Conditions

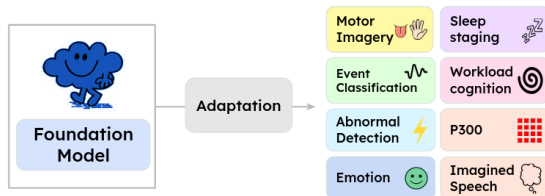
Emotion



Foundation Models for EEG Decoding

Foundation Model

Large, deep neural network pretrained in a self-supervised way (no annotations) designed to serve as the basis for various downstream tasks



Key Ingredients of EEG Foundation Models

- 1 **Tokenization** – turning continuous EEG into tokens
- 2 **Positional Encoding** – adding spatiotemporal information
- 3 **Transformer Architecture** – processing the tokens with self-attention
- 4 **Self-Supervised Objective** – what drives learning
- 5 **Large-scale Pretraining Data** – scale & diversity

⇒ **Downstream Generalization**

What actually works?

1: Tokenization

- **Fixed patching:** short (1-2 s) temporal patches (BIOT, CBraMod, REVE)
- **Vector quantization:** learns discrete codebooks (LaBraM, BrainOmni, LUNA)

Open question

Continuous patching is competitive with VQ codebooks – **added value of discretization unclear.**

2: Positional Encoding

- **Fourier**: adapted to any electrode configuration and signal length
- **Learned, asymmetric kernels** (CBraMod): long spatial / short temporal encoding, less transferable

What works

Fourier PE = robust default for montage-agnostic models (and multimodality!)

3: Transformer Architecture

- Self-Attention
- Encoder-Decoder during pretraining, Encoder-only for inference
- Model sizes vary a lot. Params: **0.39M** (BENDR) → **69M** (REVE) → **7B** (UniMind)

Specialized attention

- Criss-cross or Sparse attention (CBraMod, CSBrain)
- Linear attention (BIOT)
- State Space Model, alternative to attention (EEGMamba)

What works

No single winner – all SOTA models integrate temporal & spatial structure somehow.

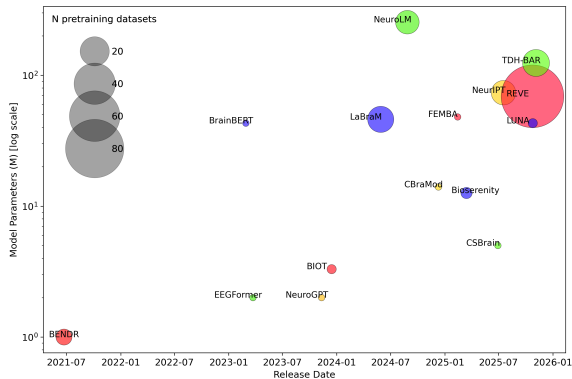
4: Self-Supervised Learning Objective

- Supervised learning: task-specific features **vs** SSL: richer, more general representations.
- **Masked autoencoding** dominates ($\sim 50\%$ masking)
- Block-wise masking $>$ random masking (REVE)
- Signal: **raw EEG** (majority) vs. **spectral** (LaBraM, Brant-2)

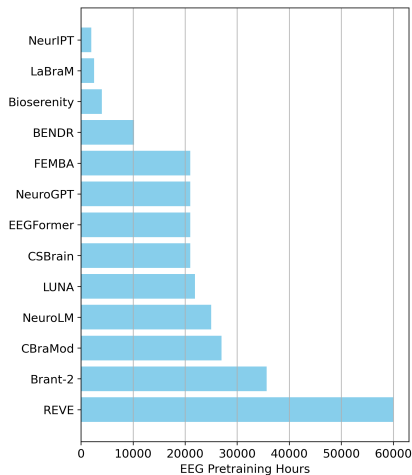
Open question

Optimal reconstruction space **not yet settled**.

5: Large-Scale, Diverse Pretraining Data

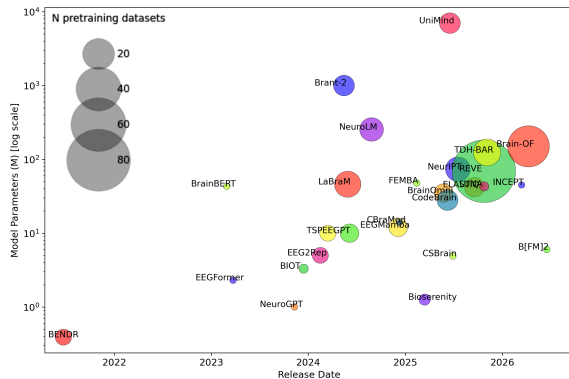


- Pretraining 2,000 $\rightarrow$ **60,000 hours (92 datasets)**
- Curation $>$ volume: 5% of data can match 100%

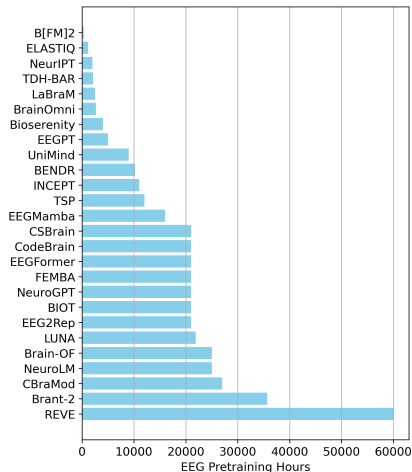


 Tang, Yuting, et al. "EEG-DLite: Dataset Distillation for Efficient Large EEG Model Training." *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. No. 21. 2026.

It Goes Really Fast



September 2026

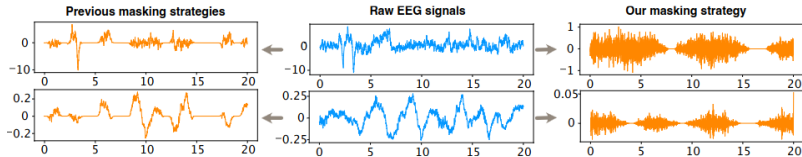


Tang, Yuting, et al. "EEG-DLite: Dataset Distillation for Efficient Large EEG Model Training." *Proceedings of the AAAI Conference on Artificial Intelligence*. Vol. 40. No. 21. 2026.

New models, new benchmarks, a few genuinely new ideas

- New **masking and reconstruction objectives**: gaussian spectral masking, flow matching
- New **critical / benchmark** papers
- New **Interpretability** work

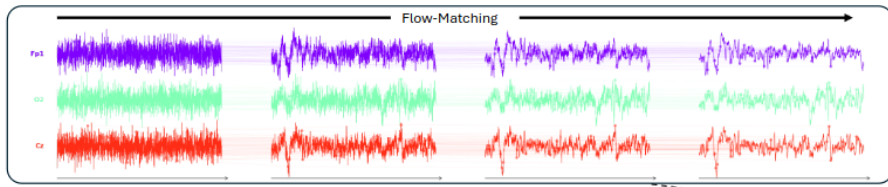
New: Gaussian-Smoothed Spectral Masking



- **Problem:** Block masking biases learning toward high frequencies
- **Idea:** Gaussian-smoothed masking on **time and frequency** (STFT)

■ Darankoum et al., *SpecMoE: Spectral Mixture-of-Experts Foundation Model for Cross-Species EEG Decoding*, [arXiv:2603.16739](https://arxiv.org/abs/2603.16739)

New: Foundation Models via Flow Matching



- **Problem:** patches/codebooks fragment continuous brain rhythms
- **Idea: flow matching** on raw waveform (no patches, no masking)
- Learn the mapping between Gaussian noise and EEG time series

Nice results

- SOTA on **7/9** tasks with only **307 hours** pretraining ($\sim 30\times$ less than REVE)
- Generated EEG fools board-certified neurologists

A Critical Look at the Field

- Little/Under investigated Scaling evidence
- **Linear probing is not enough** (full fine-tuning needed)
- $\Rightarrow$ doubts on how much SSL pretraining truly helps (Yang et al. 2026 evaluation against both classic neural network and non-neural decoders across classification and regression tasks)
- Heterogeneous and limited evaluation makes it hard to assess performances/added value

■ Kuruppu et al., *Foundation Models for Brain Signals: A Critical Review of Current Progress and Future Directions* (J. Neural Eng. 2026 Mar 10;23(2)., Yang, Liuyin, et al. "Are EEG foundation models worth it? comparative evaluation with traditional decoders in diverse BCI tasks." *The Fourteenth International Conference on Learning Representations*. 2026.

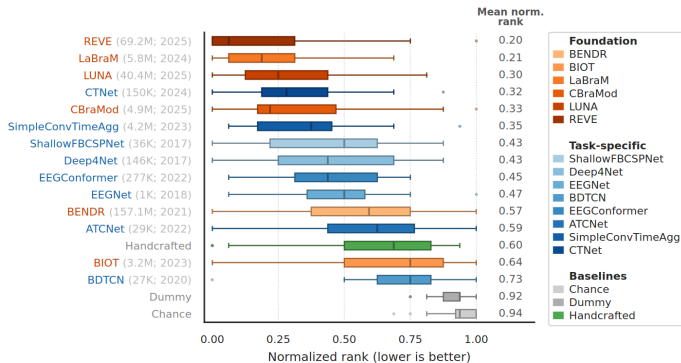
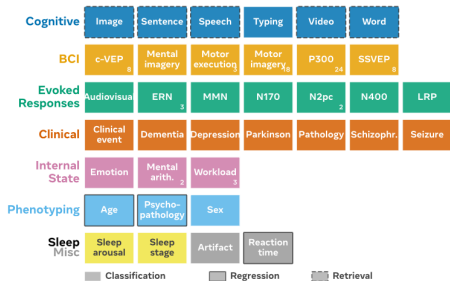
The Need for Better Benchmarks

- Fair comparison is hard: budgets, task subsets, train/test leakage
- Need for a rigorously held-out benchmark with diverse tasks, populations and evaluation schemes
- Fast-growing benchmark landscape (e.g. **NeuralBench**, OmniEEG-Bench, AdaBrain-Bench, **OpenEEG-Bench**)

■ Kuruppu et al., *Foundation Models for Brain Signals: A Critical Review of Current Progress and Future Directions* (J. Neural Eng. 2026 Mar 10;23(2)., Yang, Liuyin, et al. "Are EEG foundation models worth it? comparative evaluation with traditional decoders in diverse BCI tasks." *The Fourteenth International Conference on Learning Representations*. 2026.

NeuralBench

A Downstream tasks



Banville, Hubert, et al. "NeuralBench: A Unifying Framework to Benchmark NeuroAI Models." *arXiv preprint arXiv:2605.08495* (2026).

Take-Home Messages

- 1 EEG foundation models are moving fast
- 2 Representation + SSL are key ingredients
- 3 New paradigms are emerging beyond masked reconstruction
- 4 Better understanding of scaling laws and impact of data quality/diversity is needed
- 5 Better benchmarks are essential to prove their added value

Team & Acknowledgments



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Thank you! Questions?

Comparative Analysis: Balanced Accuracy Across Tasks

Method	TUAB	TUEV	PhyMI	BCI2a	FACED	ISRUC	Mumtaz	MAT	BCI3
EEGNet	0.764	0.388	0.581	0.448	0.489	0.713	0.901	0.659	0.395
EEGConformer	0.776	0.407	0.605	0.470	0.456	0.741	0.932	0.602	0.431
SPaRCNet	0.790	0.416	0.593	0.464	0.447	0.750	0.910	0.649	0.393
ContraWR	0.775	0.438	0.589	0.468	0.441	0.757	0.930	0.647	0.415
CNN-Transformer	0.778	0.409	0.605	0.460	0.460	0.729	0.910	0.656	0.417
FFCL	0.785	0.398	0.573	0.447	0.448	0.738	0.913	0.647	0.412
ST-Transformer	0.797	0.398	0.604	0.465	0.473	0.750	0.916	0.678	0.413
BIOT	0.778	0.435	0.594	0.478	0.673	0.709	0.668	0.623	0.609
LaBraM-Base	0.786	0.442	0.611	0.495	0.687	0.720	0.681	0.650	0.626
CBraMod	0.800	0.453	0.609	0.497	0.704	0.736	0.694	0.658	0.641
REVE-Base	0.832	0.623	0.596	0.508	0.718	0.741	0.703	0.717	0.665
B[FM]²	0.835	0.648	0.635	0.672	0.725	0.754	0.715	0.732	0.675

- **B[FM]²** (flow matching, 307h pretraining) sets a new SOTA on 7/9 tasks, ahead of REVE-Base and CBraMod across most benchmarks

Interpretability: What Do EEG FMs Actually Capture?

- **68.6%** of (model, task, feature) units are representation-causal
- Models generally attend to neurophysiologically plausible regions (motor cortex for movement tasks, occipital for visual/eyes-open-closed tasks)
- Classic EEG features are encoded somewhere in FMs. Frequency-domain features are encoded most strongly and consistently
- Most features are not only encoded but also used for inference
- on harder tasks 25~55% of the model's advantage comes from something the classic features don't capture
- Linear-probing performance often blamed on "low-quality representations" is largely a pooling artifact. Swapping mean-pooling for token-flattening pooling recovers accuracy

■ Tang, Ling, et al. "What Do EEG Foundation Models Capture from Human Brain Signals?." *arXiv preprint arXiv:2605.11410* (2026); Sirca, Urban, et al. "Beyond Accuracy: Robustness, Interpretability and Expressiveness of EEG Foundation Models." *arXiv preprint arXiv:2605.17562* (2026).